

Expectation Decider

Assume the dataset sample:

Student	Study Hours	Attendance	Group Discussion	Final Exam
S1	12	85%	Yes	Pass
S2	8	75%	No	Fail
S3	15	90%	Yes	Pass

Total Students = 3

1. Understanding the Basics

What is Probability

Probability measures the **chance that an event will occur**.

Formula

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

Key Probability Terminology

Experiment

Process that produces an outcome.

Sample Space (S)

All possible outcomes.

$$S = \{Pass, Fail\}$$

Event (E)

A specific outcome.

Example: Student passes exam.

Probability Event Examples from Dataset

Example 1: Probability of Passing

Number of students passing = 2

Total students = 3

$$P(\text{Pass}) = \frac{2}{3}$$

$$P(\text{Pass}) = 0.67$$

Example 2: Probability of High Attendance (>80%)

Students with attendance >80%

S1 = Yes

S2 = No

S3 = Yes

Total = 2

$$P(\text{Attendance} > 80) = \frac{2}{3}$$

$$= 0.67$$

Example 3: Probability of Participating in Group Discussion

Students participating = 2

$$P(\text{GroupDiscussion} = \text{Yes}) = \frac{2}{3}$$

$$= 0.67$$

2. Types of Events

Empirical Probability

Empirical probability is based on **observed data**.

Example: Passing probability from dataset.

$$\begin{aligned}P(\text{Pass}) &= \frac{2}{3} \\ &= 0.67\end{aligned}$$

Theoretical Probability

Suppose theory says **each student has equal chance to pass or fail**

Possible outcomes = {Pass, Fail}

$$\begin{aligned}P(\text{Pass}) &= \frac{1}{2} \\ &= 0.5\end{aligned}$$

3. Random Variable & Probability Distribution

Define Random Variable

Let

X = Number of students passing among 3 randomly selected students

Possible values

$$X = 0, 1, 2, 3$$

Probability Distribution Table

Using dataset probability:

$$\begin{aligned}P(\text{Pass}) &= \frac{2}{3} \\ P(\text{Fail}) &= \frac{1}{3}\end{aligned}$$

Using **Binomial Distribution**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Where

$$n = 3$$

$$p = 2/3$$

P(X=0)

$$\begin{aligned} P(0) &= \binom{3}{0} \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^3 \\ &= 1 \times 1 \times \frac{1}{27} \\ &= \frac{1}{27} = 0.037 \end{aligned}$$

P(X=1)

$$\begin{aligned} P(1) &= \binom{3}{1} \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^2 \\ &= 3 \times \frac{2}{3} \times \frac{1}{9} \\ &= \frac{6}{27} = 0.222 \end{aligned}$$

P(X=2)

$$\begin{aligned} P(2) &= \binom{3}{2} \left(\frac{2}{3}\right)^2 \left(\frac{1}{3}\right)^1 \\ &= 3 \times \frac{4}{9} \times \frac{1}{3} \\ &= \frac{12}{27} = 0.444 \end{aligned}$$

P(X=3)

$$\begin{aligned} P(3) &= \binom{3}{3} \left(\frac{2}{3}\right)^3 \left(\frac{1}{3}\right)^0 \\ &= 1 \times \frac{8}{27} \times 1 \end{aligned}$$

$$= \frac{8}{27} = 0.296$$

Probability Distribution Table

X	P(X)
0	0.037
1	0.222
2	0.444
3	0.296

Mean of Random Variable

Formula

$$\begin{aligned} \text{Mean} &= np \\ &= 3 \times \frac{2}{3} \\ &= 2 \end{aligned}$$

Variance

Formula

$$\begin{aligned} \text{Var}(X) &= np(1 - p) \\ &= 3 \times \frac{2}{3} \times \frac{1}{3} \\ &= \frac{6}{9} \\ &= 0.67 \end{aligned}$$

4. Venn Diagram Probability

Let

A = Students studying >10 hours

B = Students with attendance >80%

From dataset

$$A = \{S1, S3\}$$

$$B = \{S1, S3\}$$

Intersection

$$A \cap B = \{S1, S3\}$$

Number satisfying both = 2

5. Contingency Table

Group Discussion	Pass	Fail	Total
Yes	2	0	2
No	0	1	1
Total	2	1	3

Joint Probability

Participates AND Passes

$$\begin{aligned} P(Yes \cap Pass) &= \frac{2}{3} \\ &= 0.67 \end{aligned}$$

Marginal Probability

Probability of Passing

$$\begin{aligned} P(Pass) &= \frac{2}{3} \\ &= 0.67 \end{aligned}$$

Conditional Probability

Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Where

A = Pass

B = Group Discussion

$$\begin{aligned}P(\text{Pass} \mid \text{Discussion}) &= \frac{2/3}{2/3} \\ &= 1\end{aligned}$$

6. Understanding Relationships

Conditional Probability Interpretation

Students who **participate in group discussions** have a **100% probability of passing** in this dataset.

Independence Check

Two events are independent if

$$P(A \mid B) = P(A)$$

Here

$$\begin{aligned}P(\text{Pass} \mid \text{Discussion}) &= 1 \\ P(\text{Pass}) &= 0.67\end{aligned}$$

Since

$$1 \neq 0.67$$

Therefore events are **Dependent**.

Mutually Exclusive Check

Mutually exclusive means

$$P(A \cap B) = 0$$

But

$$P(\text{Discussion} \cap \text{Pass}) = \frac{2}{3}$$

So they are **NOT mutually exclusive**.

7. Bayes Theorem

Given

$$P(\text{Pass} \mid \text{HighAttendance}) = 0.70$$

$$P(\text{Fail} \mid \text{HighAttendance}) = 0.40$$

$$P(\text{HighAttendance}) = 0.60$$

We need

$$P(\text{Pass} \mid \text{HighAttendance})$$

Using Bayes formula

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Let

A = Pass

B = High Attendance

Step 1: Find P(Pass)

$$\begin{aligned} P(\text{Pass}) &= 1 - P(\text{Fail}) \\ &= 1 - 0.40 \\ &= 0.60 \end{aligned}$$

Step 2: Apply Bayes Formula

$$\begin{aligned}P(\text{Pass} \mid \text{HighAttendance}) &= \frac{0.70 \times 0.60}{0.60} \\&= \frac{0.42}{0.60} \\&= 0.70\end{aligned}$$

Final Result

$$P(\text{Pass} \mid \text{HighAttendance}) = 0.70$$

So, **70% probability that a student passes if attendance is high.**